Resilience Under Pressure: A Data-Driven Framework for Global Disaster Risk, Response, and Recovery Analytics

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***Abstract*—This project presents a data-driven analytical framework for quantifying global disaster resilience using multi-source datasets and computational indicators. By integrating data from EM-DAT, World Bank, UNDP, and climate sources, we construct derived metrics for disaster exposure, vulnerability, and adaptive capacity. Using feature engineering and custom mathematical models, including the Disaster Impact Index (DII), Resilience Recovery Score (RS), and Composite Resilience Index (CRI), we evaluate how countries resist and recover from disasters. An interactive Tableau dashboard visualizes trends across time, geography, and socio-economic dimensions, providing insights into which nations are at risk, which recover fastest, and which factors influence a country's resilience. The framework demonstrates how analytical modeling and visualization can guide better disaster preparedness and policy decision-making.**

***Index Terms*—data-driven, disaster resilience, risk analytics, EM-DAT, computational indicators, policy decision-making.**

1. Introduction

Natural and human-induced disasters have increased in frequency and severity over recent decades, exposing systemic weaknesses in governance, infrastructure, and social resilience all over the world. Rapid urbanization, climate change, and population growth greatly influence the impacts of floods, wildfires, pandemics, and industrial accidents. Hence, organizations require proactive, data-intelligent systems to assess risk, vulnerability, and recovery capabilities. The Global Disaster and Humanitarian Response Agency (GDHRA) commissioned this project to conceptualize and implement a prototype analytical platform capable of modeling and visualizing global disaster resilience. As resilience cannot be measured directly, the system infers it using derived indicators through feature engineering. Tableau is used to communicate insights visually for better perception through interactive dashboards.

1.1. Problem Statement and Objectives

1.1.1. Problem Statement. Global disasters such as floods, earthquakes, droughts, and epidemics continue to increase in frequency and severity. Still, resilience cannot be directly measured, and existing global dashboards do not provide integrated, data-driven exposure, recovery capacity, and vulnerability. The government and humanitarian organizations lack tools to compare countries, assess recovery speed, or identify factors influencing resilience. Available data is scattered across multiple public sources, comes in inconsistent formats, and needs to be combined for analytical modeling. There is a need for a unified analytical system that quantifies the resilience and also shows the patterns over time and geography through visualizations.

1.1.2. Project Objectives. The primary challenge is the lack of a standardized, quantifiable metric that synthesizes diverse global indicators (disaster incidence, economic health, social vulnerability) into a single, actionable measure of **disaster resilience**. Our specific objectives are:

1) To **integrate** multi-domain datasets together related to disasters, climate risks, and socio-economic conditions.
2) To **engineer** quantitative metrics, specifically the **Disaster Impact Index (DII)**, **Resilience Recovery Score (RS)**, and **Composite Resilience Index (CRI)**.
3) To **evaluate** global resilience by comparing national performance on these derived metrics across time and geography, and providing insights into global resilience patterns, and identifying relationships.
4) To **develop** an interactive Tableau dashboard allowing temporal, geographical, and comparative analysis to explore risk drivers and recovery patterns.

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2. Data Sources and Preprocessing

The analytical framework relies on the integration of several authoritative global data sources to cover the dimensions of exposure, vulnerability, and adaptive capacity.

2.1. Data Sources

Four datasets were selected for the analysis:

1) **EM-DAT (The International Disaster Database):** Provides historical data on the occurrence and impact of natural and technological disasters, including figures for deaths, injured people, and total damages.
2) **World Bank World Development Indicators:** Offers macro-economic data (e.g., GDP, GDP per capita, population, urbanization rate, governance indicators, and CO2 emissions) used to proxy a nation's financial adaptive capacity.
3) **UNDP (United Nations Development Programme):** Supplies social and human development indicators (e.g., Human Development Index, literacy rate, health indicators, and poverty rates) critical for assessing societal vulnerability.
4) **CPRI (Climate and Ecological Risk Indicators):** Contains country-level climate stress and environmental exposure metrics, including long-term drought risk, heatwave threat degree, extreme rainfall severity, ecological exposure, environmental degradation indicators, and climate vulnerability indices used to quantify climate-driven hazards.

2.2. Data Cleaning and Preprocessing

The integrated dataset was prepared through a series of systematic preprocessing steps applied to all four data sources: EM-DAT, CPRI, HDI, and World Bank Development Indicators.

2.2.1. Missing Value Handling.

- Rows missing critical identifiers such as Country or ISO codes were removed.
- Missing categorical attributes (e.g., Disaster Type, Subtype) were filled with "Unknown" to maintain completeness.
- Numeric gaps in disaster variables (deaths, affected, homeless, magnitude, damage) were imputed using type-wise medians, while zeros were used only where analytically appropriate.
- Missing climate indicators in CPRI (drought risk, heatwave threat, rainfall degree, ecological exposure) were replaced using country-wise medians to preserve geographic consistency.
- HDI and World Bank missing values were handled through time-series interpolation, median imputation, and forward/backward filling depending on the variable type.
- Remaining dataset-level gaps after merging were replaced using column medians to ensure a complete panel.

2.2.2. Standardization & Data Cleaning.

- Column names across all datasets were trimmed, cleaned, and standardized to a uniform naming scheme.
- Country names were harmonized using a unified dictionary to allow cross-dataset merging (e.g., "United States of America" to "United States").
- All relevant fields were converted into proper numeric data types.
- Rows with duplicate entries were removed.
- Non-numeric characters present in climate files and economic indicators were cleaned and cast correctly.
- Year fields were standardized to ensure a consistent Country–Year structure across all four datasets.

2.2.3. Outlier Detection & Treatment.

- Highly skewed variables such as **Total_Deaths**, **Total_Affected**, **Total_Damage**, and selected nonlinear climate risk indicators were treated using the Interquartile Range (IQR) capping method.
- Extreme growth rates or economic outliers in the World Bank data (e.g., GDP jumps, population anomalies) were capped to reduce distortion in model calculations.
- Outliers in HDI-related human development indicators were corrected or replaced based on regional trends to preserve realistic values.

2.2.4. Normalization & Scaling.

- Climate risk indicators in CPRI were log-transformed to reduce right-skewed distributions.
- After feature generation, composite climate and exposure scores were scaled using MinMax normalization for cross-country comparability.
- Final indices such as **DII**, **RRS**, and **CRI** were normalized to a 0–1 range to facilitate visualization in dashboards (e.g., Tableau).

2.2.5. Attribute Transformations & Data Reshaping.

- EM-DAT event-level records were aggregated into Country–Year summaries (frequency, severity metrics, magnitude averages, total losses).
- The World Bank dataset, originally in wide multi-year format, was reshaped into long format and then pivoted into individual indicator columns (e.g., GDP growth, literacy rate, sanitation, electricity access).
- HDI columns were cleaned, renamed, converted to numeric, and aligned with the rest of the dataset.
- A placeholder field **Disaster_Type** = "Mixed/All" was created to represent aggregated multi-disaster years.
- Cross-dataset merging was completed using standardized Country–Year keys.

2.2.6. Feature Engineering. Several new attributes were created to support resilience analysis:

- **Mortality_Rate** = deaths ÷ affected
- **Affected_Pct** = affected population ÷ total population
- **Severity_Score** and **Severity_Category**
- **Exposure** = disaster frequency × magnitude
- **Annualized_Disaster_Frequency**
- **Loss_Normalized_GDPpc** = economic loss relative to GDP per capita
- **Infrastructure_Exposure** = disaster density relative to urban population
- **Climate_Exposure_Index** (mean of drought, heat, rainfall, ecological risk)
- **Climate_Vulnerability_Index** (weighted environmental + food security + ecological stress)

The final **Composite Indices** derived were the **Disaster Impact Index (DII)**, **Resilience Recovery Score (RRS)**, and **Composite Resilience Index (CRI)**. Each engineered attribute contributes to measuring exposure, vulnerability, and adaptive capacity.

2.2.7. Final Dataset Consolidation. All four datasets (**EM-DAT**, **CPRI**, **HDI**, **World Bank**) were merged into a unified country-year panel. Final checks ensured removal of impossible values, replacement of infinities, and verification of population, GDP, and climate consistency. The cleaned dataset now serves as a multi-layered analytical framework connecting disasters, climate risks, socio-economic conditions, and development indicators for resilience modeling and dashboard visualization.

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3. Methods and Implementation

3.1. Architecture

The proposed analytical platform for disaster resilience assessment follows a data-driven, multi-layer architecture designed for integration, computation, and visualization:

1) **Data Layer:** Sources include **EM-DAT** (disasters), **World Bank** (economic indicators), **UNDP HDI**, **NASA CPRI** (climate risk), and **Our World in Data**. Its purpose is to store raw and preprocessed data in a unified country–year panel, following preprocessing steps: missing value handling, standardization, outlier treatment, normalization, and feature engineering.
2) **Computation Layer:** Implements resilience indices (**DII**, **RRS**, **CRI**) using derived features. It performs normalization and scaling to standardize metrics for cross-country comparison and integrates multi-source variables to capture exposure (**E**), vulnerability (**V**), and adaptive capacity (**A**).
3) **Visualization Layer: Tableau** dashboards provide interactive exploratory views. It supports multi-dimensional analysis across time, geography, disaster type, and socio-economic categories, utilizing visual components such as choropleth maps, scatter plots, bubble plots, time series, and treemaps.

System Flow: Data ingestion → preprocessing → feature engineering → index computation → visualization → interactive analysis.

3.2. Algorithms and Theoretical Background

The core methodology involves **feature engineering** to transform raw data into composite, explanatory indices. The custom mathematical models designed for this analysis include:

3.2.1. Disaster Impact Index (DII). The **Disaster Impact Index (DII)** measures the disaster impact relative to a country's wealth and size, combining fatalities and affected population normalized by GDP per capita and scaled by average severity.

$$DII = \frac{\text{Fatalities}_{\text{100k}} + \text{Affected}_{\text{Pct}}}{GDP_{\text{pc}}} \times \text{Avg}_{\text{Severity}}$$

Where:

- $\text{Fatalities}_{\text{100k}}$: Fatalities per 100k people.
- $\text{Affected}_{\text{Pct}}$: Affected population (percentage of total population).
- GDP_{pc}: GDP per capita.
- $\text{Avg}_{\text{Severity}}$: Average severity score (based on disaster type/magnitude).

3.2.2. Resilience Recovery Score (RRS). The **Resilience Recovery Score (RRS)** is a metric that combines the immediate economic response (GDP growth change) with underlying structural components (Human Development and Governance Effectiveness).

$$RRS = GDP^{\text{Change}}_{\text{Growth}} + \frac{HDI + Gov_{\text{Effectiveness}}}{2}$$

Where:

- $GDP^{\text{Change}}_{\text{Growth}}$: GDP growth rate change (assumed to represent post vs. pre-disaster).
- HDI: Human Development Index value (2023).
- $Gov_{\text{Effectiveness}}$: Governance Effectiveness score.

3.2.3. Composite Resilience Index (CRI). The **Composite Resilience Index (CRI)** is the final consolidated metric, representing overall resilience as the ratio of Adaptive Capacity to the combined forces of Exposure and Vulnerability.

$$CRI = \frac{\text{Adaptive}_{\text{Capacity}}}{\text{Exposure} \times \text{Vulnerability}}$$

Where the components are calculated as:

- $\text{Adaptive}_{\text{Capacity}} = \frac{HDI + \text{Adult}_{\text{Literacy}} + Gov_{\text{Effectiveness}}}{3}$
- $\text{Exposure} = \text{Annualized}_{\text{Frequency}} \times \text{Avg}_{\text{Severity}}$
- $\text{Vulnerability} = \text{Fatalities}_{100\text{k}} + \text{Loss}_{\text{Normalized GDPpc}}$

A higher CRI indicates greater resilience.

3.2.4. Index Computation Algorithm.

- Merge datasets by **Country–Year** keys.
- Compute feature-level metrics: mortality rate, affected population percentage, disaster frequency, magnitude, economic loss per GDP.
- Calculate derived indices: **DII**, **RRS**, and **CRI**.
- Normalize all indices to **0–1** for visualization and comparability.

3.2.5. Comparative Analysis Algorithm.

- Scatter plots: Exposure vs. Vulnerability, CRI vs. GDP per capita, CRI vs. Governance Effectiveness.
- Quadrant analysis to classify countries: High/Low Exposure & High/Low Vulnerability.
- Temporal grouping: **HDI**-based categorization to track resilience evolution.

3.2.6. Feature Engineering Algorithm.

- Compute **Mortality_Rate**, **Affected_Pct**, **Severity_Score**, **Annualized_Disaster_Frequency**, **Loss_Normalized_GDPpc**, **Infrastructure_Exposure**, **Climate_Exposure_Index**, and **Climate_Vulnerability_Index**.
- Aggregate event-level data into country-year summaries.
- Scale and normalize features to ensure analytical consistency.

3.3. Implementation Details (Tools and Libraries)

The data manipulation and analytical modeling were primarily performed using **Python** with libraries such as **Pandas** and **NumPy** for data handling and **Scikit-learn** for modeling and **PCA**. The final output visualization was developed using an **interactive Tableau dashboard**.

3.4. Implementation Workflow

The implementation workflow follows a structured pipeline from raw data to interactive insights:

1) **Data Ingestion:** Collect open datasets from multiple sources and align data on a Country–Year basis.
2) **Preprocessing:** Handle missing values, standardize fields, and detect and cap outliers.

TABLE 1. TOOLS AND LIBRARIES USED FOR IMPLEMENTATION

Component	Tool / Library	Purpose
Data Cleaning and Preprocessing	Python (pandas, numpy)	Merging, imputing missing values, normalization, feature engineering
Computation and Index Calculation	Python	Derivation of DII, RRS, CRI indices
Visualization and Dashboard	Tableau Desktop	Interactive maps, scatter plots, bubble charts, time series, treemaps
Data Sources	EM-DAT, World Bank, UNDP HDI, NASA CPRI, Our World in Data	Multi-source integration for comprehensive resilience modeling

3) **Feature Engineering:** Derive exposure, vulnerability, and adaptive capacity metrics; aggregate event-level data; and compute normalized features.
4) **Index Computation:** Calculate **DII**, **RRS**, and **CRI** for each country-year, and normalize indices.
5) **Visualization and Dashboard:** Build multi-view **Tableau** dashboards, configure interactive filters (Year, Region, Disaster Type, HDI Group), and implement dual-axis plots, scatter/bubble plots, and choropleth maps.
6) **Analytical Storytelling:** Identify countries with high exposure but low vulnerability, analyze resilience drivers, and explore temporal trends.

Outcome: The platform provides a dynamic, interactive, and comparative view of global disaster resilience, supporting decision-makers with actionable insights for policy and risk management.

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4. Results and Interpretation

The analytical framework developed in this study integrates disaster exposure, socio-economic vulnerability, and adaptive capacity to quantify national resilience. Using the **Composite Resilience Index (CRI)**, **Resilience Recovery Score (RRS)**, and **Disaster Impact Index (DII)**, combined with multi-source data fusion and Tableau-based visualization, the results reveal distinct global patterns of risk, resilience, and inequality.

4.1. Key Results

4.1.1. Identifying Nations With High Exposure but Low Vulnerability.

A logarithmic scatter plot of Average **Exposure** versus Average **Vulnerability** was used to classify nations through a four-quadrant analytical model. The quadrant of interest—Quadrant IV (**High Exposure–Low Vulnerability**)—represents the world's most **resilient countries in relation to geographic risk**. The analysis identifies nations such as **Australia**, **Canada**, **Malaysia**, **Saudi**

Arabia, **Panama**, and **Uruguay** as outliers demonstrating strong resilience performance. These countries face high hazard frequency or intensity but exhibit vulnerability levels significantly below the global mean, indicating effective governance, robust early-warning systems, and resilient infrastructure.

4.1.2. Economic Wealth vs. Governance: Which Drives Resilience?. Comparative scatter plots of **CRI** vs. **GDP per capita** and **CRI** vs. **Governance Effectiveness** were analyzed. Results indicate that while **GDP** exhibits a moderate positive association with **CRI**, **Governance Effectiveness** reveals a stronger and more consistent positive trend. This demonstrates that **governance quality serves as the more reliable determinant of national resilience**, as institutional strength enhances disaster planning, health infrastructure, and emergency response capabilities.

4.1.3. Temporal Evolution of Resilience and Climate Risk. A dual-axis time-series visualization plotting average **Exposure** and average **CRI** for High, Medium, and Low **HDI** groups revealed significant divergence:

- For **Low HDI nations**, the slope of **Exposure** increases more steeply than **CRI**, indicating a **widening resilience gap** where climate risk is rising faster than adaptive capacity.
- **High HDI nations** maintain a stable or improving resilience trend, with adaptive capacity growth equal to or exceeding the growth of exposure, keeping their resilience gap closed.

This confirms that the world's poorest nations face a deteriorating resilience trajectory, underscoring an urgent need for targeted investment.

4.2. Visualizations and Interpretation

4.2.1. Logarithmic Scatter Plot: Exposure vs. Vulnerability. To address the question of effective risk mitigation—identifying nations that successfully decouple high geographic risk from high socio-economic vulnerability—a logarithmic scatter plot was constructed comparing Average Log_Exposure against Average Log_Vulnerability. This analysis utilized a four-quadrant model where the Bottom-Right Quadrant (Quadrant IV) clearly identifies these resilient nations as statistical outliers that provide key benchmarks for effective disaster risk reduction policies.

A calculated filter was applied to isolate countries strictly falling into Quadrant IV, providing a concise list of nations demonstrating this strong resilience performance. The resulting list, detailed in Figure 10, includes countries like Australia, Canada, and Saudi Arabia, confirming their status as high-exposure, low-vulnerability outliers, and establishing them as key examples for effective governance and resilient infrastructure.

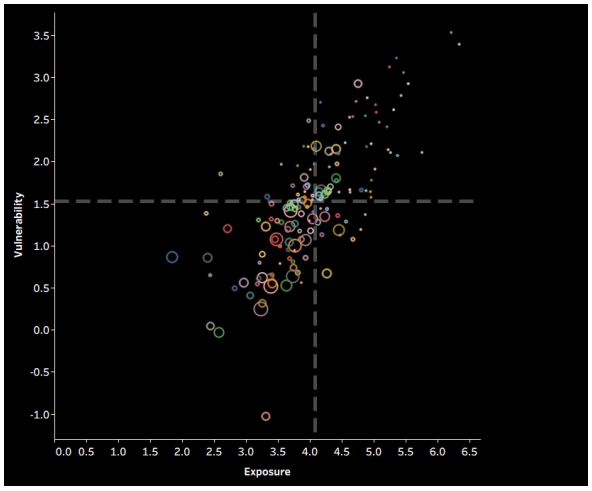
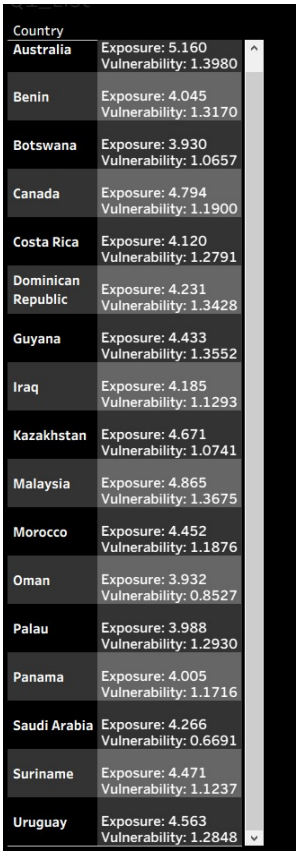


Figure 1. Logarithmic Scatter Plot: Average Log Exposure vs. Average Log Vulnerability with Four-Quadrant Classification. The size of the bubble represents the Composite Resilience Index (CRI).



Country	
Australia	Exposure: 5.160 Vulnerability: 1.3980
Benin	Exposure: 4.045 Vulnerability: 1.3170
Botswana	Exposure: 3.930 Vulnerability: 1.0657
Canada	Exposure: 4.794 Vulnerability: 1.1900
Costa Rica	Exposure: 4.120 Vulnerability: 1.2791
Dominican Republic	Exposure: 4.231 Vulnerability: 1.3428
Guyana	Exposure: 4.433 Vulnerability: 1.3552
Iraq	Exposure: 4.185 Vulnerability: 1.1293
Kazakhstan	Exposure: 4.671 Vulnerability: 1.0741
Malaysia	Exposure: 4.865 Vulnerability: 1.3675
Morocco	Exposure: 4.452 Vulnerability: 1.1876
Oman	Exposure: 3.932 Vulnerability: 0.8527
Palau	Exposure: 3.988 Vulnerability: 1.2930
Panama	Exposure: 4.005 Vulnerability: 1.1716
Saudi Arabia	Exposure: 4.266 Vulnerability: 0.6691
Suriname	Exposure: 4.471 Vulnerability: 1.1237
Uruguay	Exposure: 4.563 Vulnerability: 1.2848

Figure 2. Filtered List of Resilient Nations (Quadrant IV) with Log Exposure and Log Vulnerability Scores.

4.2.2. Comparative Scatter Plots: CRI vs. Wealth and Governance. The relative importance of wealth (GDP per Capita) versus institutional quality (Gov Effectiveness) was

determined by comparing two scatter plots against the CRI. While the plot of CRI vs. GDP per capita showed a moderate correlation, the plot of CRI vs. Governance Effectiveness revealed a stronger and more consistent positive trend. This confirmed that **institutional quality** is the more reliable predictor of national resilience outcomes.

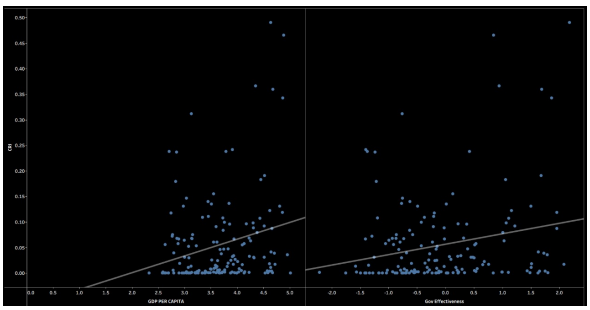


Figure 3. Comparative Scatter Plots of Composite Resilience Index (CRI) vs. GDP per Capita and CRI vs. Governance Effectiveness.

4.2.3. Dual-Axis Time-Series Plot: Resilience Evolution and the Gap. To analyze the temporal dynamics of risk, a dual-axis time-series visualization was used, plotting average Exposure and average CRI for High, Medium, and Low HDI groups. The comparison of the slopes of the lines revealed a critical divergence: the steeper slope of Exposure for Low HDI nations confirms a **widening resilience gap** where risk is outpacing adaptive capacity, underscoring severe global inequality.

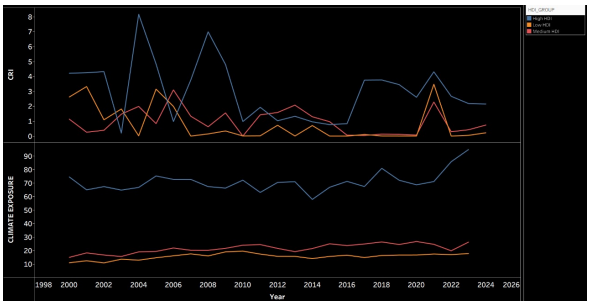


Figure 4. Dual-Axis Time-Series Visualization of Exposure and CRI for High, Medium, and Low HDI groups (1998-2022).

4.2.4. Integrated Spatial Understanding Through the Dynamic Global Resilience Map. To complement the quantitative findings, a dynamic **choropleth map** was developed to visualize the spatial distribution of CRI, RRS, and DII across countries. The map transitions between measuring outcomes (CRI), policy inputs (RRS), and disaster consequences (DII).

Comparative spatial analysis of RRS and DII serves as a strategic diagnostic tool. Countries demonstrating high RRS (strong systems) but also high DII (high impact) suggest that mitigation investments may be insufficient or overwhelmed by escalating hazards, highlighting where systems fail to translate into reduced suffering.

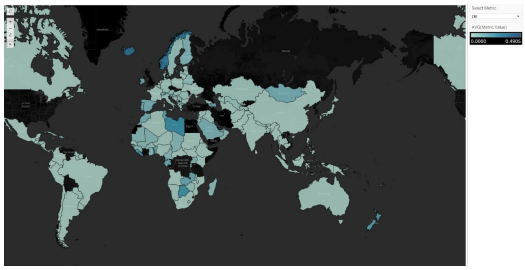


Figure 5. *
Global Map: CRI (Outcome)

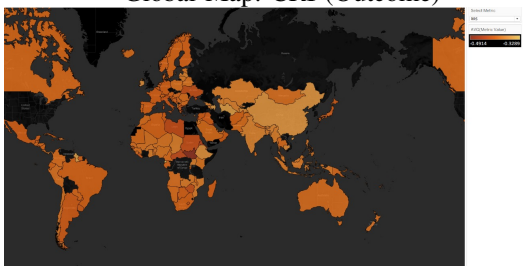


Figure 6. *
Global Map: RRS (Input)

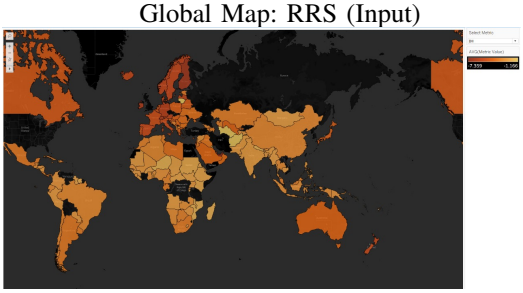


Figure 7. *
Global Map: DII (Consequence)

Figure 8. Dynamic Choropleth Maps of Global Resilience Metrics: These three maps illustrate the spatial distribution of outcomes (CRI), policy inputs (RRS), and disaster consequences (DII).

4.2.5. Disaster Impact Breakdown: The DII Treemap. A Treemap provides a hierarchical summary of the global disaster impact (DII). The **size of each rectangle** represents the total aggregate DII for a specific Region-Disaster Type combination, while the **color intensity** (Red/Blue) reflects the Average Fatalities per 100k (human cost). This allows for a quick assessment of which disaster scenarios carry the highest total impact and the highest associated human vulnerability.

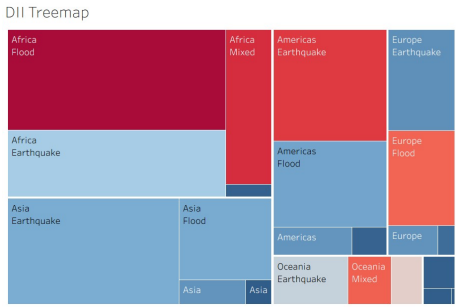


Figure 9. DII Treemap: Total Aggregate Impact (Size) by Region and Disaster Type, with Color reflecting Average Fatalities per 100k (Human Cost).

4.2.6. Overall Conclusions. Across all visualizations and analytical perspectives, several conclusions emerge:

- Resilience depends more on **governance than wealth**. Strong institutions are the most consistent predictor of high **CRI**.
- Global inequality is shaping the resilience gap. **Low HDI nations are falling further behind** as climate hazards escalate.
- High exposure does not imply high vulnerability. Strategic investments can counteract unavoidable geographic risk.
- Policy effectiveness must be evaluated against actual impact. Comparing **RRS** and **DII** highlights where systems fail to translate into reduced suffering.

Together, these findings provide a comprehensive data-driven understanding of global disaster resilience, offering clear pathways for strengthening adaptive capacity and reducing future risk.

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5. Limitations and Future Work

5.1. Limitations

- **Data Availability and Granularity:** Reliance on officially reported disaster figures (**EM-DAT**) means data quality and completeness vary significantly across countries, potentially biasing the **DII** and **RRS** for nations with poor reporting infrastructure.
- **Index Weighting Subjectivity:** The construction of the **DII** and **CRI** involves weighted sums, meaning the specific weights assigned to variables (e.g., w_1, w_2, w_3 in **DII**) are subject to domain expert judgment and may not universally represent the relative importance of impact/vulnerability factors.
- **Causal Inference:** While strong correlations were found (e.g., governance vs. **CRI**), the model primarily establishes association rather than definitive causal links between policy inputs and resilience outcomes.

5.2. Future Work

- **Predictive Modeling and Climate Projections:** Integrating climate models and machine learning techniques to forecast future exposure and vulnerability under various climate change scenarios, allowing for forward-looking risk assessment.
- **Near Real-Time Response Analysis:** Incorporating unstructured data sources like social media, satellite imagery, and localized sensors to measure immediate disaster response effectiveness and recovery trajectory at finer geographic resolution than country-level.
- **Event-Level Recovery Analysis:** Developing a sophisticated **RRS** model that tracks the recovery of specific sectors (e.g., agriculture, tourism, energy) post-disaster, rather than relying solely on aggregate macro-economic indicators.

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6. Conclusion

This study developed a comprehensive, data-driven analytical platform to measure and compare disaster resilience across countries using multi-source global datasets. By integrating **EM-DAT** disaster records with socio-economic, governance, and climate indicators from the **World Bank**, **UNDP**, **NASA**, and **Our World in Data**, the system established a unified computational representation of exposure, vulnerability, and adaptive capacity. Three core indices—the **Disaster Impact Index (DII)**, **Resilience Recovery Score (RRS)**, and **Composite Resilience Index (CRI)**—were formulated to quantify the disruptive impact of disasters, the speed and quality of post-disaster recovery, and the overall structural resilience of nations. **Tableau** dashboards enabled interactive exploration across regions, income groups, and time periods, providing clear visual narratives that highlight global inequality in resilience outcomes. The results demonstrate that **governance quality is a stronger predictor of resilience than economic wealth**, illustrating that institutional effectiveness, not income alone, determines how nations prepare for, absorb, and recover from shocks. High-exposure nations such as **Australia**, **Canada**, **Malaysia**, and **Uruguay** showed strong resilience, proving that exposure does not automatically translate into vulnerability. Conversely, **low-HDI nations** face rising climate threats without proportional improvements in adaptive capacity, widening the global resilience gap. The platform successfully meets the **GDHRA's** objective by providing a computationally transparent, visually compelling, and policy-relevant framework for understanding disaster risk and recovery. While limitations exist, the analytical structure provides a strong foundation for future enhancements, including predictive modeling, climate projections, and event-level recovery analysis.

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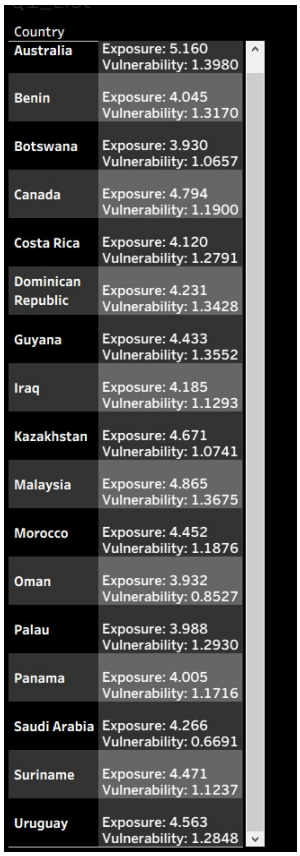
Appendix: Supplemental Visualizations



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Guyana	Exposure: 4.433 Vulnerability: 1.3552
Iraq	Exposure: 4.185 Vulnerability: 1.1293
Kazakhstan	Exposure: 4.671 Vulnerability: 1.0741
Malaysia	Exposure: 4.865 Vulnerability: 1.3675
Morocco	Exposure: 4.452 Vulnerability: 1.1876
Oman	Exposure: 3.932 Vulnerability: 0.8527
Palau	Exposure: 3.988 Vulnerability: 1.2930
Panama	Exposure: 4.005 Vulnerability: 1.1716
Saudi Arabia	Exposure: 4.266 Vulnerability: 0.6691
Suriname	Exposure: 4.471 Vulnerability: 1.1237
Uruguay	Exposure: 4.563 Vulnerability: 1.2848

Figure 10. Filtered List of Resilient Nations (Quadrant IV) with Log Exposure and Log Vulnerability Scores.

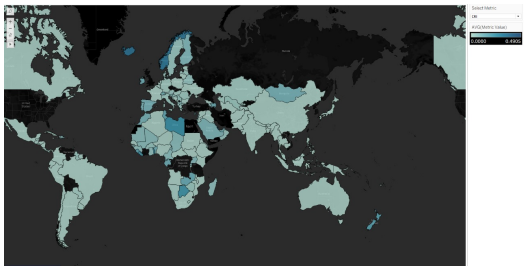

Figure 11. *
Global Map: CRI (Outcome)

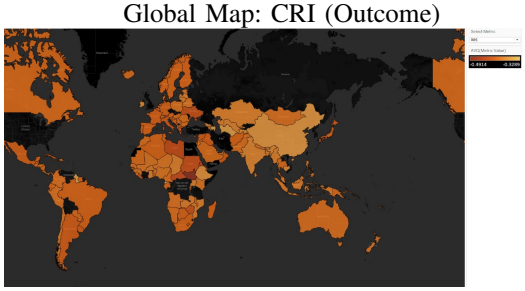

Figure 12. *
Global Map: RRS (Input)

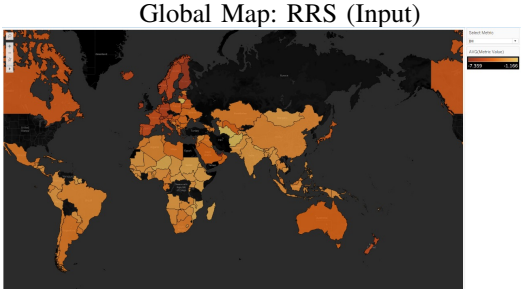

Figure 13. *
Global Map: DII (Consequence)

Figure 14. Dynamic Choropleth Maps of Global Resilience Metrics: These three maps illustrate the spatial distribution of outcomes (CRI), policy inputs (RRS), and disaster consequences (DII).